

OligoTox-Kidney — Narrative Document

NIH/NCATS Oligonucleotide Toxicity (OligoTox) Open Data Challenge, Phase 2 (Data Generation) Endpoint: kidney toxicity / nephrotoxicity

Phase 2 narrative deliverable. Target length ≤ 12 pages when rendered to PDF.

1. Executive summary

OligoTox-Kidney is an openly-licensed, per-measurement dataset of oligonucleotide **kidney toxicity**. It contains **246 measurements across 65 unique oligonucleotides** spanning **35 target genes**, every row traceable to a named source table, figure or label section, and every row strict-kidney (`is_kidney_specific = TRUE` on 246/246).

The dataset is an **in-silico curation** of published and public-domain data. No wet-lab work was performed. Its contribution is not new bench measurement but **harmonisation**: values that exist today only as scattered patent tables, label sections and supplementary files, reduced to one schema with one grading rubric and per-row provenance, so they can be trained on and audited.

Composition

subject_class	rows	share
animal_invitro (rat primary PTEC)	81	32.9%
human_invitro (primary human PTEC, PTEC-TERT1, ciPTEC, 3D-RPTEC, tubule-on-chip)	67	27.2%
animal_invivo (rat, mouse, monkey)	56	22.8%
human_clinical (trials, labels, case reports)	42	17.1%

109 of 246 rows (44.3%) are human, and **67 are human in-vitro** — the category the Phase 2 announcement names as of particular interest. A further 81 rows are animal in-vitro on the *same assay and compounds*, which is what makes the human↔animal extrapolation analysis in §5 possible.

Controls

We distinguish **designed controls** from **negative observations**, because conflating them overstates the evidence:

- **Designed negative control.** Patent compound **1-1** (OLG045), a scrambled-sequence ASO the source explicitly designates *innocuous* and uses as its normalisation reference. It appears in 25 rows and grades **0 in every one**. Two independent grading rubrics were *anchored* on it (§4.3), so it functions as a control in the strict sense.
- **Designed positive anchors.** **SPC5001** / patent compound **3-1** (OLG002/OLG047) — a known human nephrotoxin that caused acute tubular necrosis in a clinical trial — and patent compound **4-1** (OLG048), the source's *high-toxicity* reference. **Inotersen** (OLG001) anchors the severe clinical end (grade 3, crescentic glomerulonephritis).
- **Negative observations, not controls.** The remaining grade-0 rows are compounds measured at a non-toxic exposure. They are evidence, but they were not *designed* as controls, and §3 explains why some of them are weaker evidence than they appear.

Headline numbers

- 65 oligonucleotides · 246 measurements · 35 target genes · 15 bridging oligos
- Sequences for **55/65**; the 10 gaps are structural, not clerical (§4.4)
- Grade distribution 0/1/2/3 = **97 / 58 / 60 / 31** — a genuinely graded label, not a rare-event problem

- Modalities: ASO gapmer 40 · GalNAc-siRNA 12 · splice-switching ASO 4 · PMO 4 · siRNA 2 · 1st-gen PS-DNA 2 · aptamer 1
- Licence **CC BY 4.0**, with per-row redistribution status retained

2. Main findings and conclusions

2.1 Animal-to-human translation is bidirectional, not one-way

The received wisdom is that animal renal toxicology **over-predicts** human effects for 2'-MOE ASOs. Our 15 bridging oligos — those carrying evidence on both sides of the human/animal divide — do not support that as a rule:

concordance	oligos
concordant	7
animal over-predicts	6
animal under-predicts	2

The two under-predictions are the safety-relevant direction and involve the dataset's most severe human findings: **inotersen** (human grade 3 crescentic glomerulonephritis against animal grade 1) and **givosiran** (human 2, animal 1). A screening cascade calibrated on the assumption that animals over-predict would have under-called both.

This is a bounded finding, and we state the bounds: 15 oligos is not a rule, and 3 of the 6 over-prediction verdicts rest on human grade-0 values that direct source retrieval could not support (§3). Model the direction as an open question, not a constant.

2.2 Human cell systems can be more sensitive than the animal in-vivo grade

Patent compound **3-1** — the same molecule as SPC5001, which caused acute tubular necrosis and AKI in humans — grades **3 in human in-vitro** against **2 in rat in-vivo** within a single laboratory's data. The human cell system recovered severity the animal in-vivo grading did not. That is exactly the translational claim the Challenge is interested in, demonstrated on a compound whose human outcome is independently documented.

2.3 A provenance/outcome confound, quantified — and partly corrected

The most important methodological finding is about the dataset itself. When clinical rows were cross-tabulated by **how well sourced they were** against **what they concluded**, provenance predicted outcome almost perfectly: of 20 rows derived from search summaries, **zero** reached grade ≥ 2 , against 11 of 22 anchor-sourced rows (one-sided Fisher $p = 4.5 \times 10^{-10}$; expected 5.6 severe among the unverified block, observed 0).

Direct retrieval of the underlying documents for 7 of the unverified absence claims found **only one** that survived as a measured negative. The rest were cases where renal endpoints were never measured, or were not reported in the cited source — and in two instances the trials had *excluded* renally impaired patients at enrolment, so the study design could not have detected nephrotoxicity.

A dataset that encodes this without marking it teaches a model to predict "does this compound have a regulatory paper trail", not renal biology — and that shortcut inverts on prospective compounds, which have no dossier and would be scored non-toxic by construction. That is the wrong error direction for a safety model.

Two corrections were applied. First, three approved drugs with *measured* human negatives were added (§4.2), weakening the association **3.7×** to $p = 1.65 \times 10^{-10}$. Second, and more importantly, the schema now carries `renal_endpoints_measured` (§4.5), which separates "measured and unremarkable" from "never looked". **13 grade-0 clinical rows are explicitly flagged as not supported as measured negatives**. Consumers can exclude, down-weight or impute them; they can no longer be mistaken for safety evidence.

We report this prominently rather than quietly fixing it, because the confound is not unique to us — it is a structural hazard of curating safety data from heterogeneous sources, and the field would benefit from it being named.

2.4 Nephrotoxicity here is functional before it is cytotoxic

145 of 246 rows are injury-biomarker readouts and only 7 are viability. This reflects the underlying biology: phosphorothioate ASOs accumulate in proximal tubule epithelium via megalin/cubilin endocytosis and produce reversible low-molecular-weight proteinuria **without loss of viability**. A viability-only screen under-calls this phenotype. The schema therefore captures KIM-1, EGF/EGFR, α 1-microglobulin, clusterin, cystatin C and lysosomal load alongside function.

3. How the data were produced

3.1 Design

- **Grain.** One row = oligo $\times$ system/subject $\times$ delivery $\times$ concentration/dose $\times$ readout. The same oligo at one concentration read on KIM-1 *and* viability is two rows.
- **Two normalised tables**, joined on `oligo_id`: `oligos.csv` (identity and design predictors, 65 rows $\times$ 20 columns) and `measurements.csv` (outcomes and context, 246 $\times$ 25). A denormalised analysis view (`oligotox_kidney_merged.csv`, 246 $\times$ 44) is **generated, never hand-edited**.
- **Strict-kidney scope.** Verified, not asserted: 246/246 rows `is_kidney_specific=TRUE`, tissue values only kidney (79) and proximal_tubule (167), zero hepatic readouts, models or sources.

3.2 Acquisition paths

Each row records which path produced it via `source_id`:

- 1 **Local full-text extraction** of primary-source PDFs (drisapersen/ciPTEC, Sandelius urinary biomarker panel, Moisan human PTEC panel, two Roche nephrotoxicity patents).
- 2 **Secondary/review extraction**, used for cross-checking rather than as primary evidence.
- 3 **Search-summary derivation (ws)**, from an early session whose network policy blocked full-text fetch. These 36 rows are flagged and are the subject of §2.3.
- 4 **WHO INN nomenclature derivation** — described in §4.4, and the methodological contribution we would most want reused.

3.3 Computational processing

Every transformation is a committed, re-runnable script; none of the derived values is hand-entered.

`split_human_animal.py` derives `subject_class` and the bridge view; `extract_patent_table2.py` and `extract_n4_table5.py` parse the two patent tables; `fill_inn_sequences.py` derives sequences from INN nomenclature; `add_endpoint_provenance.py` derives `renal_endpoints_measured`; `build_merged.py` regenerates the analysis view.

Two extraction hazards are worth recording because they silently produce plausible wrong answers:

- **Table text layers interleave columns.** Both patent tables extract as a flat number stream that cannot be mapped back to (compound, concentration, system, timepoint). Both extractors therefore parse the *layout-preserving* text and **check parsed values against anchors taken from the printed table** before writing; both refuse to write a partial table.
- **Case encodes chemistry.** In gapmer sequences, upper case marks 2'-MOE/cEt wings and lower case the DNA gap. A case-sensitive validator reports these correct rows as malformed — ours flagged 22 valid rows before being corrected.

3.4 A corrected error, disclosed

During final QC, **21 rows — 19% of the then-dataset** — were found to record a rat study as a mouse study. The source patent binds its Table 1 to a method section describing "Wistar Han CrI : WI (Han) male rats", dosing "at 40 mg/kg on days 1 and 8", a "Multiplex MAP **Rat** Kidney Toxicity Magnetic Bead Panel 2", and sacrifice "on day 15". The rows said mouse, 7 days, dose unknown. All four statements were verified against the patent before correction. Species distribution moved from mouse 30 / rat 8 to **rat 29 / mouse 9**, and 21 rows gained a published

dose.

We disclose this because it is the kind of error that survives peer review when a dataset is presented only as totals, and because it is an argument for the per-row provenance discipline that caught it.

4. Indicators, predictors, and their distributions

4.1 Dependent variable — nephrotox_grade (0–3)

Grade	Definition	n
0	No renal signal at tested exposure	97
1	Mild/functional, reversible; no viability loss	58
2	Moderate — clear injury-biomarker rise and/or histopathology	60
3	Severe — AKI, glomerulonephritis, renal failure, dose-limiting	31

Distribution by subject class:

class	g0	g1	g2	g3
human_clinical	21	10	6	5
human_invitro	33	12	17	5
animal_invitro	37	21	12	11
animal_invivo	6	15	25	10

All grades carry grade_provisional pending subject-matter sign-off.

4.2 Readouts

injury_biomarker 145 · functional 35 · clinical_renal_outcome 30 · histopathology 27 · viability 7 · accumulation 2. Delivery is gymnotic_free_uptake 148 · systemic_dose 93 · intrathecal 3 · intravitreal 1 · oral 1. **207 of 246 rows carry a numeric dose or concentration.**

A caveat that applies to five compounds: all three DMD PMO labels warn that **serum creatinine is unreliable in Duchenne patients** because of reduced skeletal muscle mass. Creatinine-based renal readouts in DMD populations must be read with that in mind; the rows added from those labels record cystatin C, UPCR and urine dipstick instead.

4.3 Grading rubrics for the quantitative panels

The two patent panels report continuous biomarker values, so grades were thresholded. The thresholds are **stated and anchored, not free**: in each case the floor is set so that the source's own innocuous reference compound grades 0 on every one of its cells.

- **Extracellular EGF, % of saline** (human PTEC / PTEC-TERT1): <200 / 200–499 / 500–1499 / ≥1500 → 0/1/2/3. Compound 1-1 grades 0 on all 16 cells, and grade then tracks the source's own in-vivo classification monotonically (innocuous 16×0; medium 10/3/2/1; high 4/4/4/4).
- **EGFR mRNA and KIM-1, % of compound 1-1** (rat primary PTEC): KIM-1 rises with injury (<200 / 200–499 / 500–1499 / ≥1500); EGFR falls (>70 / 40–70 / 20–39 / <20). Compound 1-1 grades 0 on all 9 cells.

The two panels use **different normalisations** — % saline versus % compound 1-1 — recorded in readout_unit and flagged in notes so they are never silently pooled.

4.4 Design predictors

Variable	Distribution
Modality	ASO gapmer 40 · GalNAc-siRNA 12 · SSO 4 · PMO 4 · siRNA 2 · PS-DNA 2 · aptamer 1
Backbone	full-PS 45 · PS/PO-mix 15 · PMO-neutral 4 · mixed 1
Conjugate	none 48 · GalNAc 16 · PEG 1
Stage	research panel 30 · approved 19 · phase 3 (incl. discontinued) 9 · phase 2 5 · phase 1 1 · class-level 1
Sequence available	55 / 65
Identity confirmation	patent sequence listing 25 · WHO INN nomenclature 20 · regulatory label 7 · publication 3 · not established 10

Sequence recovery from WHO INN nomenclature. 20 sequences were recovered by parsing the residue-by-residue chemical nomenclature WHO publishes for named oligonucleotides (2'-O-methyl-P-thiocytidylyl-(3'→5')-...). This is a deterministic parse rather than a transcription, and it is self-checking in two ways: INN writes one strand with (3'→5') linkages and its partner with (5'→3'), so mishandling direction yields a *reversed* strand — the parser was validated by reproducing two sequences already in the table character-for-character before any new row was written — and every duplex was required to show its guide strand as the exact reverse complement of its sense strand, with residue counts cross-checked against the published molecular formula where the phosphorus count fixes the length. We believe this method is reusable by any group needing sequences for INN-named oligonucleotides.

The 10 remaining gaps are structural: 6 proprietary research compounds whose sponsors never published sequences, 2 class-level aggregate rows for which a single sequence is not meaningful, and 2 Ionis development-code compounds that never received an INN.

4.5 Provenance variables

renal_endpoints_measured — measured_and_reported 233 · cannot_determine 8 · not_measured 3 · not_reported_in_source 2. This field exists so that a grade of 0 on a row that is not measured_and_reported reads as **"not established"**, never as "safe".

purity_pct and purity_method are TBD **for all 65 oligos**. This is a genuine and unavoidable limitation of a curation-type dataset, and we verified it rather than assuming it: both source patents were searched for purity, HPLC, UPLC, LC-MS and mass-spectrometry language and neither reports any; labels and trial publications do not publish per-batch purity. What *can* be answered is the identity half of the requirement, and identity_confirmation records for every oligo how its identity was established.

5. The gap this addresses

Public oligonucleotide toxicity data is dominated by **hepatotoxicity**. Kidney data is thinner, more scattered, and disproportionately locked in formats that resist reuse — patent example tables, label sections, supplementary PDFs. Three specific gaps are closed here.

A harmonised kidney corpus with per-row provenance. 246 measurements under one schema and one rubric, each traceable to a named table or section, spanning in-vitro human and animal cell systems, animal in-vivo, and human clinical outcomes.

Human in-vitro data at usable scale, paired with animal data on the same compounds. 67 human in-vitro rows and 81 animal in-vitro rows, of which the patent panels supply the same compounds in **human cells, rat cells, and rat in vivo from a single laboratory** — three-way comparisons with no cross-study confounding. This is the configuration the Challenge's extrapolation criterion asks for, and it is rare in public data.

Two reusable methods. The INN sequence-recovery procedure (§4.4), and the renal_endpoints_measured distinction (§4.5). The second is the more transferable: any curated safety dataset assembled from heterogeneous sources is exposed to the same confound, and most do not carry a field that would reveal it.

6. Use in predictive modelling

6.1 What the dataset supports today

The merged view carries design predictors and graded outcome on the same row, so no join is required.

Reasonable first targets: ordinal prediction of nephrotox_grade from chemistry and design; binary "reaches grade ≥ 2 "; and — the most interesting — **predicting the human grade from the animal grade** on the bridge set.

6.2 How it must be used, to avoid learning the wrong thing

These are not generic caveats; they follow from findings in this dataset.

- 1 **Filter on renal_endpoints_measured.** Rows that are not measured_and_reported should be excluded or treated as missing-label, never as negatives. Otherwise the model learns provenance (§2.3).
- 2 **Group splits by oligo_id, never random row splits.** Compounds contribute up to 25 rows each; random splitting puts the same molecule on both sides and inflates performance.
- 3 **Do not pool the two normalisations.** % saline and % compound 1-1 are different scales; readout_unit distinguishes them.
- 4 **Treat subject_class as a first-class feature or a stratifier.** Grade distributions differ sharply across the four classes — animal in-vivo is 6/15/25/10, human clinical 21/10/6/5. A model blind to it will confound assay severity with compound severity.
- 5 **Expect small-n on the bridge.** 15 oligos supports descriptive translation analysis, not a fitted species-correction model.

6.3 Honest limits

- Grades are **provisional** pending subject-matter sign-off.
- Purity is absent for every oligo (§4.5).
- 36 WS rows remain search-derived; 13 grade-0 clinical rows are flagged unsupported.
- The confound is **reduced, not eliminated** ($p = 1.65 \times 10^{-11}$).
- 39 rows still lack a numeric dose, mostly awaiting FDA Pharmacology/Toxicology reviews.

We would rather ship a dataset whose weaknesses are enumerated and machine-readable than one whose totals look cleaner. Every limitation above is either a flagged column value or a documented row set, so a modeller can act on it programmatically rather than having to rediscover it.

7. Access

All data, schema, extraction scripts and documentation are in the project repository under toxicity/kidney/, licensed **CC BY 4.0** (repository-root LICENSE), with per-row redistribution status retained for values whose underlying sources carry their own terms. Source PDFs are included for verification and are not covered by that licence.

Artefact	Path
Oligo design table	data/oligos.csv (65 × 20)
Measurement table	data/measurements.csv (246 × 25)
Merged analysis view	data/oligotox_kidney_merged.csv (246 × 44, generated)
Human/animal bridge view	data/human_animal_bridge.csv (15 oligos)
Data dictionary & schema	schema.md
Methodology	METHODOLOGY.md (long form), METHODOLOGY_PHASE2.md (submission)
Validation of the confound	CLINICAL_VALIDATION.md
Independent strict review	REVIEW-2026-08.md
Public access plan	PADP.md

